Tutorial 3

Q1. What are the differences of FETs and BJTs?

Q2: Explain the construction and operation of a n/p-channel JFET.

Q3. Draw the characteristics curve of a JFET and explain.

Q4. What is polarization? How polarization can create a channel in a E-MOSFET?

Q5. What is fixed/self /voltage divider biasing? Draw the load line and Q point of a fixed biased JFET amplifier. How you can adjust the Q point of the fixed biased JFET amplifier?

Q6. Draw a small signal AC model for an N-channel JFET in CS connection and explain all the values.

Q7. Find out the voltage gain expression of a fixed/self/voltage divider biased JFET amplifier in CS connection.

Question 1:

For the JFET circuit in Figure 1, $V_{GS(off)}$ = -4 V and I_{DSS} = 12 mA. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation when V_{GS} = 0 V.

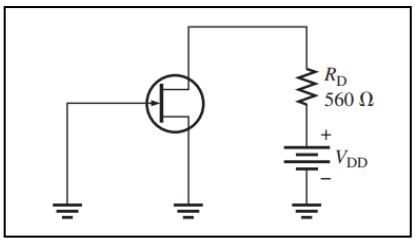


Figure 1

Question 2

a) A typical JFET has the following parameters: I_{DSS} = 9 mA and $V_{GS(off)}$ = -8 V (maximum). Using these values, determine the drain current for V_{GS} = 0 V, -1 V and 4 V.
b) A typical JFET has the following parameters: I_{DSS} = 3.0 mA, $V_{GS(off)}$ = -6 V maximum, and $g_{fs(max)}$ = 5000 μS. Using these values, determine the forward transconductance for V_{GS} = -4 V, and find I_D at this point
c) A certain JFET has an I_{GSS} of -2 nA for V_{GS} = -20 V. Determine the input resistance

Question 3:

a) Find V_{DS} and V_{GS} in the JFET circuit in Figure 2.

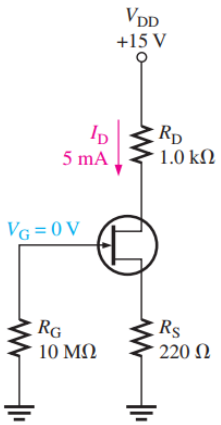


Figure 2

b) Determine I_D and V_{GS} for the JFET with voltage-divider bias in the circuit in Figure 3, given that for this particular JFET the parameter values are such that $V_D \approx 7$ V.

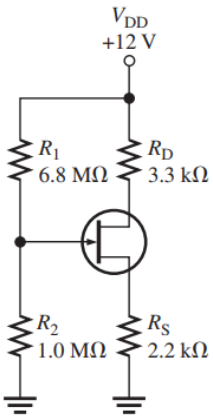


Figure 3

Question 4

a) A typical E-MOSFET gives $I_{D(on)}$ = 500 mA (minimum) at V_{GS} = 10 V and $V_{GS(th)}$ = 1 V. Determine the drain current for V_{GS} = 5 V.

b) For a certain D-MOSFET, I_{DSS} = 10 mA and $V_{GS(off)}$ = -8 V. Is this an *n*-channel or a *p*-channel D-MOSFET? Calculate I_D at V_{GS} = -3 V. Calculate I_D at V_{GS} = +3 V.

Question 5:

a) What is the total output voltage for the unloaded amplifier in Figure 4? I_{DSS} is 4.3 mA and $V_{GS(off)}$ is -2.7 V. If a 4.7 kΩ resistor is AC coupled to the output of the amplifier below, what is the resulting output voltage?

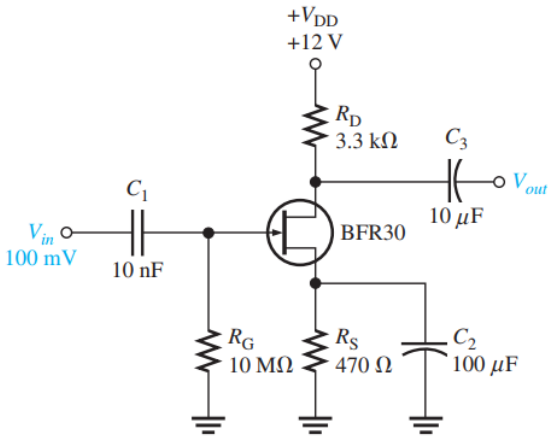


Figure 4

b) What is the input resistance seen by the signal source in the circuit in Figure 5? I_{GSS} = 30 nA at V_{GS} = 10 V.

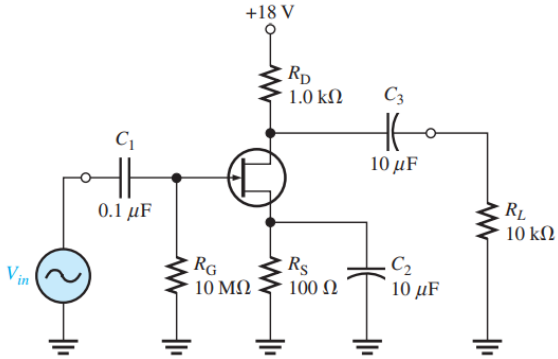


Figure 5

Question 6:

Find V_{GS}, I_D, V_{DS} and the AC output voltage for the circuit in Figure 6. Assume that $I_{D(on)}$ = 200 mA at V_{GS} = 4 V, $V_{GS(th)}$ = 2 V, g_m = 23 mS. V_{in} = 25 mV.

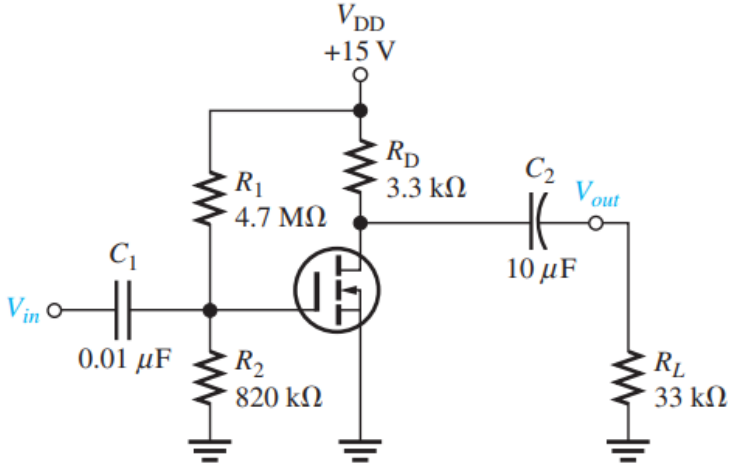


Figure 6